Homework Solutions: Sections 5.3 & 5.5

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Section 5.3

Problem 5.3-3

Given: Independent random variables X_1 and X_2 with PDFs:

$$f_1(x_1) = 2x_1, \quad 0 < x_1 < 1$$
$$f_2(x_2) = 4x_2^3, \quad 0 < x_2 < 1$$

(a) Compute $P(0.5 < X_1 < 1$ **and** $0.4 < X_2 < 0.8)$

Since X_1 and X_2 are independent, the joint probability is the product of the marginal probabilities.

$$P(0.5 < X_1 < 1) = \int_{0.5}^{1} 2x_1\,dx_1 = \left[x_1^2\right]_{0.5}^{1} = 1^2 - (0.5)^2 = 1 - 0.25 = 0.75$$
$$P(0.4 < X_2 < 0.8) = \int_{0.4}^{0.8} 4x_2^3\,dx_2 = \left[x_2^4\right]_{0.4}^{0.8} = (0.8)^4 - (0.4)^4 = 0.4096 - 0.0256 = 0.384$$
$$P(\text{Joint}) = 0.75 \times 0.384 = \mathbf{0.288}$$

(b) Compute $E(X_1^2X_2^3)$

Due to independence, $E(X_1^2X_2^3) = E(X_1^2) \cdot E(X_2^3)$.

$$E(X_1^2) = \int_0^1 x_1^2(2x_1)\,dx_1 = \int_0^1 2x_1^3\,dx_1 = \left[\frac{2x_1^4}{4}\right]_0^1 = \frac{1}{2}$$
$$E(X_2^3) = \int_0^1 x_2^3(4x_2^3)\,dx_2 = \int_0^1 4x_2^6\,dx_2 = \left[\frac{4x_2^7}{7}\right]_0^1 = \frac{4}{7}$$
$$E(X_1^2X_2^3) = \frac{1}{2} \cdot \frac{4}{7} = \mathbf{\frac{2}{7}}$$

Problem 5.3-6

Given: A random sample of size $n = 2$ from a distribution with PDF $f(x) = 6x(1-x)$ for $0 < x < 1$. Let $Y = X_1 + X_2$.

First, we find the mean and variance of the individual variable X.

$$E(X) = \int_0^1 x(6x - 6x^2)\,dx = \int_0^1 (6x^2 - 6x^3)\,dx = \left[2x^3 - \frac{3}{2}x^4\right]_0^1 = 2 - 1.5 = 0.5$$
$$E(X^2) = \int_0^1 x^2(6x - 6x^2)\,dx = \int_0^1 (6x^3 - 6x^4)\,dx = \left[\frac{6}{4}x^4 - \frac{6}{5}x^5\right]_0^1 = 1.5 - 1.2 = 0.3$$
$$Var(X) = E(X^2) - [E(X)]^2 = 0.3 - (0.5)^2 = 0.05$$

Now, find the mean and variance of Y.

$$E(Y) = E(X_1) + E(X_2) = 0.5 + 0.5 = \mathbf{1}$$
$$Var(Y) = Var(X_1) + Var(X_2) = 0.05 + 0.05 = \mathbf{0.10}$$

Problem 5.3-11

Given: Independent variables $X_1 \sim b(4, 1/2)$, $X_2 \sim b(6, 1/3)$, $X_3 \sim b(12, 1/6)$.

(a) $P(X_1 = 2, X_2 = 2, X_3 = 5)$

$$P(X_1 = 2) = \binom{4}{2}(0.5)^4 = 0.375$$
$$P(X_2 = 2) = \binom{6}{2}\left(\frac{1}{3}\right)^2\left(\frac{2}{3}\right)^4 \approx 0.3292$$
$$P(X_3 = 5) = \binom{12}{5}\left(\frac{1}{6}\right)^5\left(\frac{5}{6}\right)^7 \approx 0.0284$$
$$P(\text{Joint}) = 0.375 \times 0.3292 \times 0.0284 \approx \mathbf{0.0035}$$

(b) $E(X_1X_2X_3)$ For Binomial distributions, $\mu = np$.

$$E(X_1) = 4(0.5) = 2, \quad E(X_2) = 6(1/3) = 2, \quad E(X_3) = 12(1/6) = 2$$

$$E(X_1X_2X_3) = 2 \cdot 2 \cdot 2 = \mathbf{8}$$

(c) Mean and Variance of $Y = X_1 + X_2 + X_3$ For Binomial distributions, $\sigma^2 = np(1-p)$.

$$\begin{aligned} E(Y) &= 2 + 2 + 2 = \mathbf{6} \\ Var(Y) &= Var(X_1) + Var(X_2) + Var(X_3) \\ &= 4(0.5)(0.5) + 6\left(\frac{1}{3}\right)\left(\frac{2}{3}\right) + 12\left(\frac{1}{6}\right)\left(\frac{5}{6}\right) \\ &= 1 + \frac{4}{3} + \frac{5}{3} = 1 + \frac{9}{3} = \mathbf{4} \end{aligned}$$

Section 5.5

Problem 5.5-2

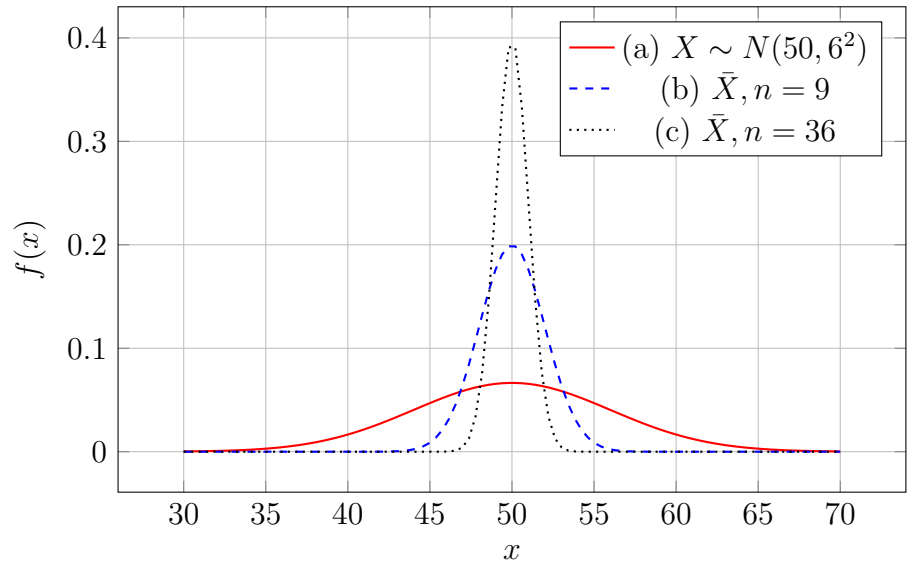
Given: $X \sim N(50, 36)$. Therefore $\mu = 50, \sigma = 6$. We compare the PDFs of:

(a) X: $\sigma = 6$.

(b) $\bar{X}$ $(n = 9)$: $\sigma_{\bar{x}} = \frac{6}{\sqrt{9}} = 2$.

(c) $\bar{X}$ $(n = 36)$: $\sigma_{\bar{x}} = \frac{6}{\sqrt{36}} = 1$.

Comparison of PDFs for Problem 5.5-2



Problem 5.5-8

Given: Male $X \sim N(184.09, 39.37)$ and Female $Y \sim N(171.93, 50.88)$. We seek $P(X > Y)$, or $P(X - Y > 0)$. Let $W = X - Y$.

$$\mu_W = \mu_X - \mu_Y = 184.09 - 171.93 = 12.16$$
$$\sigma_W^2 = \sigma_X^2 + \sigma_Y^2 = 39.37 + 50.88 = 90.25$$
$$\sigma_W = \sqrt{90.25} = 9.5$$

We calculate the Z-score for $W = 0$:

$$Z = \frac{0 - 12.16}{9.5} = -1.28$$

$$P(X > Y) = P(W > 0) = P(Z > -1.28) = 1 - P(Z < -1.28) \approx \mathbf{0.8997}$$

Problem 5.5-10

Given: $\mu = 800, \sigma = 100$. We want smallest n such that sum S_n satisfies $P(S_n \geq 10,000) \geq 0.90$. $S_n \sim N(n\mu, n\sigma^2) = N(800n, 10000n)$. Standardizing:

$$P\left(Z \geq \frac{10,000 - 800n}{100\sqrt{n}}\right) = 0.90$$

The critical Z-value for the lower tail of 0.10 is -1.282.

$$\frac{10,000 - 800n}{100\sqrt{n}} \leq -1.282$$

$$100 - 8n \leq -1.282\sqrt{n} \implies 8n - 1.282\sqrt{n} - 100 \geq 0$$

Solving the quadratic for $\sqrt{n}$ yields $\sqrt{n} \approx 3.616$, so $n \approx 13.08$. Rounding up to the next integer:

$$\mathbf{n = 14}$$

Problem 5.5-15

Given: $T \sim t(17)$.

(a) $t_{0.01}(17)$ Looking at the t-table for $\alpha = 0.01$ and $df = 17$: **2.567**.

(b) $t_{0.95}(17)$ By symmetry, $t_{0.95} = -t_{0.05}$. Looking up $\alpha = 0.05$, $df = 17$ gives 1.740. Result: **-1.740**.

(c) $P(-1.740 \leq T \leq 1.740)$ Since 1.740 is the critical value for $\alpha = 0.05$, the area in the two tails is $2(0.05) = 0.10$. The area in the middle is $1 - 0.10 = \mathbf{0.90}$.